Nama: Nindira adiarty

NIM: 2510514006

Mata Kuliah: Matematika Sains Data

Dosen Pengampu: Muhammad Panji Muslim, S.Pd., M.Kom.

Parameter Personalisasi

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NIM: 2510514006 (Nindira adiarty)

Simbol	Definisi	Rumus	Nilai
N	Dua digit akhir	2510514006 mod 100	N = 06
a	Digit puluhan NIM	floor (06/10)	a = 0
b	Digit satuan NIM	06 mod 10	b = 6
K	Konstanta skala	(a + b + 1) = (0 + 6 + 1)	K = 7
θ_0 (dimapat)	Sudut awal bil. kompleks	(a · 30) + (b · 5)	$\theta_0 = 30°$
α	Learning rate gradient	0,001 × (b + 1)	α = 0,007

Soal 2 Nindira adiarty 2510514006

Matriks kunci A = $\begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix}$ nilai $a = 0$, $b = 6$

Maka subs a dan b = $\begin{bmatrix} 1 & 0 & 2 \\ 1 & 2 & 6 \\ 6 & 1 & 3 \end{bmatrix}$

a) Hitung det (A) dengan ekspansi kofaktor baris 1.

$$\det(A) = 1 \times \begin{bmatrix} 2 & 6 \\ 1 & 3 \end{bmatrix} - 0 \times \begin{bmatrix} 1 & 6 \\ 6 & 3 \end{bmatrix} + 2 \times \begin{bmatrix} 1 & 2 \\ 6 & 1 \end{bmatrix}$$

$$= 1.(6-6) - 0 + 2.(1-12)$$

$$= 1.0 - 0 + 2.(-11)$$

$$= -22$$

b) Hitung M_{ij} untuk semua $i, j \in \{1,2,3\}$

i/j	j=1	j=2	j=3
i=1	$M_{11} = \begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix} = 6-6=0$	$M_{12} = \begin{pmatrix} 1 & 6 \\ 6 & 3 \end{pmatrix} = 3-36=33$	$M_{13} = \begin{pmatrix} 1 & 2 \\ 6 & 1 \end{pmatrix} = 1-12=-11$
i=2	$M_{12} = \begin{pmatrix} 0 & 2 \\ 1 & 3 \end{pmatrix} = 0-2=-2$	$M_{22} = \begin{pmatrix} 1 & 2 \\ 6 & 3 \end{pmatrix} = 3-12=-9$	$M_{23} = \begin{pmatrix} 1 & 0 \\ 6 & 1 \end{pmatrix} = 1-0=1$
i=3	$M_{13} = \begin{pmatrix} 0 & 2 \\ 2 & 6 \end{pmatrix} = 0-4=-4$	$M_{32} = \begin{pmatrix} 1 & 2 \\ 1 & 6 \end{pmatrix} = 6-2=4$	$M_{33} = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} = 2-0=2$

c) Susun matriks kofaktor C

$$C = \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} = \begin{bmatrix} 0 & 33 & -11 \\ 2 & -9 & -1 \\ -4 & -4 & 2 \end{bmatrix}$$

Matriks adjoin $Adj(A) = C^T = \begin{bmatrix} 0 & 2 & -4 \\ 33 & -9 & -4 \\ -11 & -1 & 2 \end{bmatrix}$

Hitung $A^{-1} = \left(\frac{1}{det(A)}\right) \cdot Adj(A)$ bentuk pecahan eksak.

$$A^{-1} = \frac{1}{-22} \begin{bmatrix} 0 & 2 & -4 \\ 33 & -9 & -4 \\ -11 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & +2/22 & -4/22 \\ +33/22 & -9/22 & -4/22 \\ -11/22 & -1/22 & 2/22 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1/11 & 2/11 \\ -3/2 & 9/22 & 2/11 \\ 1/2 & 1/22 & -1/11 \end{bmatrix}$$

e) Blok piksel asli

$$P = \begin{bmatrix} 120 & (50+N) & 200 \\ 80 & a+2 & (10+N) \\ (30+N) & 90 & 170 \end{bmatrix}$$

dengan

$N = 6$

$a = 0$

Enkripsi $E = A.P$ * * *

$$E = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 2 & 6 \\ 6 & 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 120 & 56 & 200 \\ 80 & 2 & 16 \\ 36 & 90 & 170 \end{bmatrix}$$

$E_{11} = (1 \times 120) + (0.80) + (2.36) = 120 + 0 + 72 = 192$

$E_{12} = (1.56) + (0 + 2) + (2.90) = 56 + 0 + 180 = 236$

$E_{13} = (1.200) + (0 \times 16) + (2.170) = 200 + 0 + 340 = 540$

} Baris 1

$E_{21} = (1.120) + (2.80) + (6 \times 36) = 120 + 160 + 216 = 496$

$E_{22} = (1 \cdot 56) + (2.2) + (6.90) = 56 + 4 + 540 = 600$

$E_{23} = (1.200) + (2.16) + (6.170) = 200 + 32 + 1020 = 1252$

} Baris 2

$E_{31} = (6.120) + (1 \times 80) + (3.36) = 720 + 80 + 108 = 908$

$E_{32} = (6.56) + (1.2) + (3.90) = 336 + 2 + 270 = 608$

$E_{33} = (6.200) + (1 \times 16) + (3.170) = 1200 + 16 + 510 = 1726$

} Baris 3

Maka matriks $E = \begin{bmatrix} 192 & 236 & 540 \\ 496 & 600 & 1252 \\ 908 & 608 & 1726 \end{bmatrix}$

Deskripsi $P' = A^{-1} . E$ ***

$$Adj(A) . E = \begin{bmatrix} 0 & 2 & -4 \\ 33 & -9 & -4 \\ -11 & -1 & 2 \end{bmatrix} \begin{bmatrix} 192 & 236 & 540 \\ 496 & 600 & 1252 \\ 908 & 608 & 1726 \end{bmatrix}$$

Maka $M = Adj(A) . E$

$M_{11} = (0.192) + (2.496) + (-4.908) = 0 + 992 - 3632 = -2640$

$M_{12} = (0.236) + (2.600) + (-4.608) = 0 + 1200 - 2432 = -1232$

$M_{13} = (0.540) + (2.1252) + (-4.1726) = 0 + 2504 - 6904 = -4400$

Baris 1

$M_{21} = (33.192) + (-9.496) + (-4.908) = 6336 - 4464 - 3632 = -1760$

$M_{22} = (33.236) + (-9.600) + (-4.608) = 7788 - 5400 - 2432 = -44$

$M_{23} = (33.540) + (-9.1252) + (-4.1726) = 17820 - 11268 - 6904 = -352$

Baris 2

$M_{31} = (-11.192) + (-1.496) + (2.908) = -2112 - 496 + 1816 = -792$

$M_{32} = (-11.236) + (-1.600) + (2.608) = -2596 - 600 + 1216 = -1980$

$M_{33} = (-11.540) + (-1.1252) + (2.1726) = -5940 - 1252 + 3452 = -3740$

Baris 3

Lalu $$P^{-1} = \frac{1}{-22} \begin{bmatrix} -2640 & -1232 & -4400 \\ -1760 & -44 & -352 \\ -792 & -1980 & -3740 \end{bmatrix}$$

$$= \begin{bmatrix} -2640/-22 & -1232/-22 & -4400/-22 \\ -1760/-22 & -44/-22 & -352/-22 \\ -792/-22 & -1980/-22 & -3740/-22 \end{bmatrix} = \begin{bmatrix} 120 & 56 & 200 \\ 80 & 2 & 16 \\ 36 & 90 & 170 \end{bmatrix}$$

Maka $P' = P$, Artinya deskripsi sempurna.

Jika $\det(A) = 0$, maka tidak ada invers, $P' = A^{-1} . E$ tidak bisa dihitung, matriks singular, citra terenkripsi tidak dapat dikembalikan ke asalnya.